

ASSIGNMENT 01

GIT, GITEA, GITHUB, GIT LFS & GITHUB PAGES

BS(SE)-V Cloud Computing (BSE-410)

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TASK 1 — Install Gitea and Push a Repository

1.1 Install and Run Gitea

Step 1.1: Install and Run Gitea

Installed Docker Engine and Docker Compose on the Ubuntu server. Cloned the official Gitea Docker repository, configured the `compose.yaml` file, initialized PostgreSQL along with the Gitea container service, and verified local connectivity via HTTP status code 200 on port 3000.



Library



Type here to search

My Computer

Ubuntu 64-bit

Ubuntu 64-bit

My Computer

System information as of Mon Oct 5 12:10:25 PM UTC 2026

```
System load: 0.01      Processes:      223
Usage of /: 52.8% of 9.75GB  Users logged in: 0
Memory usage: 25%      IPv4 address for ens33: 192.168.201.129
Swap usage: 0%
```

* Canonical Workshop gives developers fast, composable, reproducible, and secure developer environments that are perfect for agentic workflows.

<https://ubuntu.com/workshop>

Expanded Security Maintenance for Applications is not enabled.

62 updates can be applied immediately.
34 of these updates are standard security updates.
To see these additional updates run: `apt list --upgradable`

Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: `sudo pro status`

```
laibaamjad065@ubuntu:~$ git clone https://github.com/WaqasSaleem97/Gitea.git
Cloning into 'Gitea'...
```

```
remote: Enumerating objects: 16, done.
remote: Counting objects: 100% (16/16), done.
remote: Compressing objects: 100% (9/9), done.
remote: Total 16 (delta 4), reused 7 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (16/16), 5.37 KiB | 5.37 MiB/s, done.
Resolving deltas: 100% (4/4), done.
```

```
laibaamjad065@ubuntu:~$ cd Gitea
```

```
laibaamjad065@ubuntu:~/Gitea$ docker compose up -d
```

```
[+] up 16/20
```

```
[+] up 25/25
```

```
✓ Image gitea/gitea:latest Pulled
✓ Image postgres:17-alpine Pulled
✓ Network webnet Created
✓ Volume gitea_postgres Created
✓ Volume gitea Created
✓ Container gitea Started
✓ Container gitea_db Started
```

```
laibaamjad065@ubuntu:~/Gitea$
```

```
laibaamjad065@ubuntu:~/Gitea$ docker compose ps
```

NAME	IMAGE	COMMAND	SERVICE	CREATED
gitea	gitea/gitea:latest	"/usr/bin/entrypoint..."	gitea	14 seconds ago
gitea_db	postgres:17-alpine	"docker-entrypoint.s..."	db	14 seconds ago

```
laibaamjad065@ubuntu:~/Gitea$ curl -sS -o /dev/null -w "Gitea HTTP status: %{http_
```

```
Gitea HTTP status: 200
```

```
laibaamjad065@ubuntu:~/Gitea$
```

1.2 Create and Push Assignment 01 to Gitea

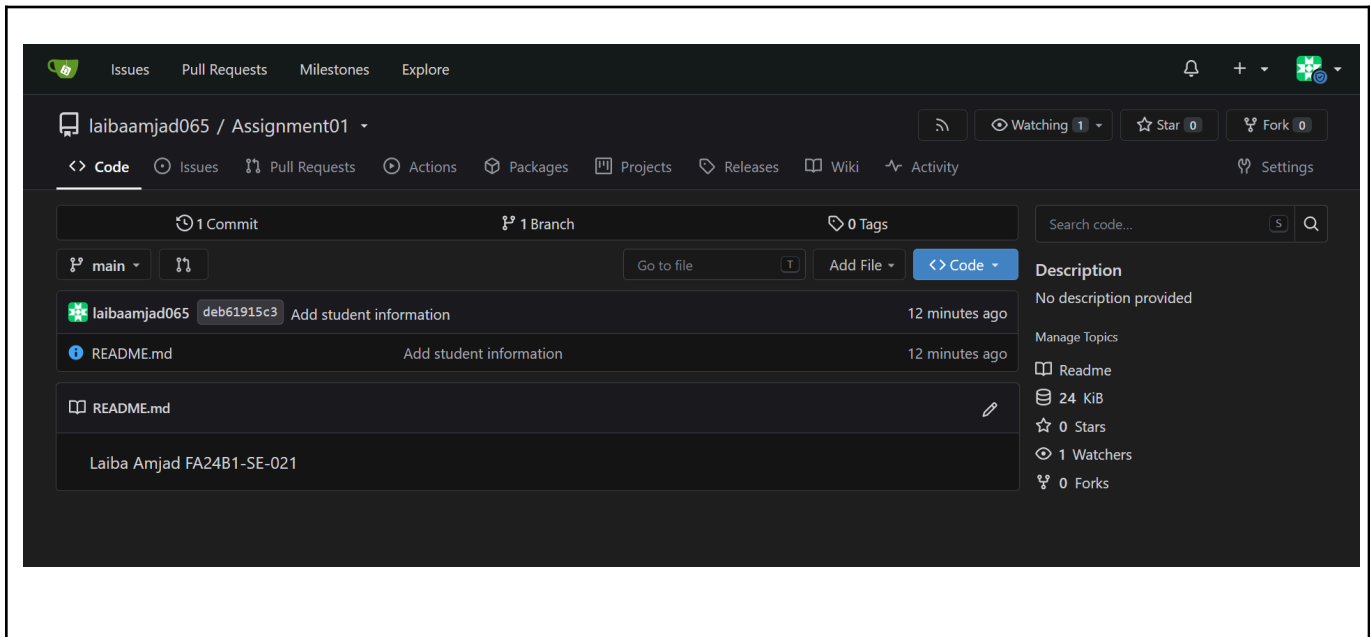
Created a local repository named **Assignment01** on the Ubuntu server containing a **README.md** file with student registration details. Generated a Gitea Personal Access Token (**ubuntu-server-token**), added the local Gitea remote URL without embedding credentials, and successfully pushed the **main** branch over HTTP using token authentication.

```
laibaamjad065@ubuntu:~/Gitea$ cd ~
laibaamjad065@ubuntu:~$ mkdir Assignment01
laibaamjad065@ubuntu:~$ cd Assignment01
laibaamjad065@ubuntu:~/Assignment01$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to 'main' in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/laibaamjad065/Assignment01/.git/
laibaamjad065@ubuntu:~/Assignment01$ git branch -M main
laibaamjad065@ubuntu:~/Assignment01$ echo -e "Laiba Amjad\nFA24B1-SE-021" > README.md
laibaamjad065@ubuntu:~/Assignment01$ git add README.md
laibaamjad065@ubuntu:~/Assignment01$ git commit -m "Add student information"
[main (root-commit) deb6191] Add student information
1 file changed, 2 insertions(+)
create mode 100644 README.md
laibaamjad065@ubuntu:~/Assignment01$ git remote add gitea http://laibaamjad065@127.0.0.1:3000/laibaamjad065/Assignment01.git
laibaamjad065@ubuntu:~/Assignment01$ git remote -v
gitea  http://laibaamjad065@127.0.0.1:3000/laibaamjad065/Assignment01.git (fetch)
gitea  http://laibaamjad065@127.0.0.1:3000/laibaamjad065/Assignment01.git (push)
laibaamjad065@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://laibaamjad065@127.0.0.1:3000':
remote: Failed to authenticate user
fatal: Authentication failed for 'http://127.0.0.1:3000/laibaamjad065/Assignment01.git/'
laibaamjad065@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://laibaamjad065@127.0.0.1:3000':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 252 bytes | 63.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To http://127.0.0.1:3000/laibaamjad065/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'gitea/main'.
laibaamjad065@ubuntu:~/Assignment01$
```

1.3 Verify Gitea Repository

Verified the public **Assignment01** repository in the Gitea web interface. Confirmed that the repository ownership, visibility status (Public), and rendered **README.md** content displaying student identification details are accurate.

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TASK 2 — Push the Same Repository to GitHub

2.1 Add GitHub Remote

Created a public GitHub repository named **Assignment01**. Added the GitHub remote URL to the existing local repository on the Ubuntu server and verified that both **gitea** and **github** remotes are configured correctly without exposed passwords or access tokens.

```
laibaamjad065@ubuntu:~/Assignment01$ git remote -v
gitea    http://laibaamjad065@127.0.0.1:3000/laibaamjad065/Assignment01.git (fetch)
gitea    http://laibaamjad065@127.0.0.1:3000/laibaamjad065/Assignment01.git (push)
github   https://github.com/laibaamjad065/Assignment01.git (fetch)
github   https://github.com/laibaamjad065/Assignment01.git (push)
laibaamjad065@ubuntu:~/Assignment01$
```

2.2 Push to GitHub

Pushed the local **main** branch to the GitHub remote repository using SSH/HTTPS authentication.

```

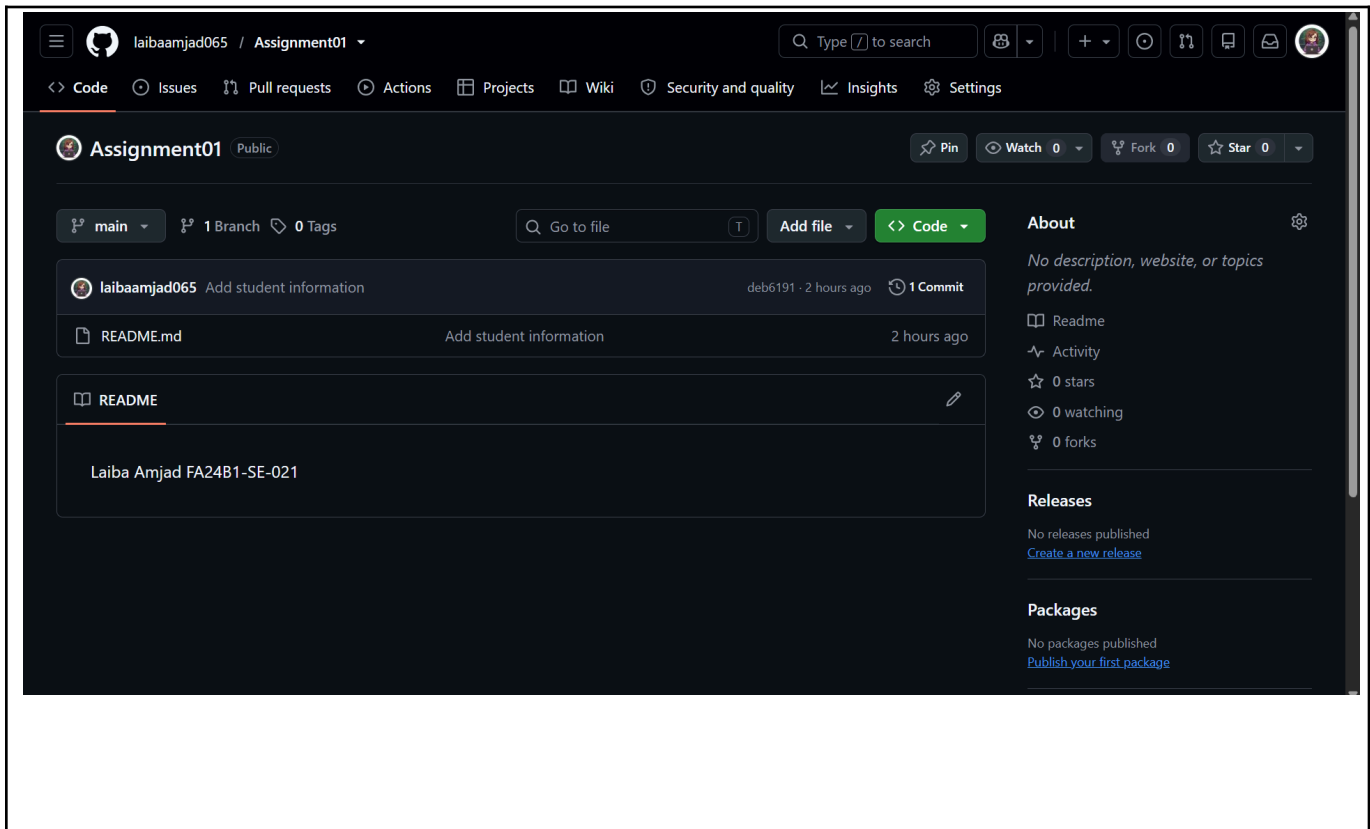
fatal: Authentication failed for 'https://github.com:laibaamjad065/Assignment01.git'
laibaamjad065@ubuntu:~/Assignment01$ ssh-keygen -t ed25519 -C "fa24b1-se-021@student.domain"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/laibaamjad065/.ssh/id_ed25519):
/home/laibaamjad065/.ssh/id_ed25519 already exists.
Overwrite (y/n)? y
Enter passphrase for "/home/laibaamjad065/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/laibaamjad065/.ssh/id_ed25519
Your public key has been saved in /home/laibaamjad065/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:7gT1a2oMnPGmoCMECGiXE7HhKh2q0Eymhk0AYRLN/c fa24b1-se-021@student.domain
The key's randomart image is:
+--[ED25519 256]--+
|0+oo+o|
|B=+++.o|
|+=+o.o .|
|Bo . oE .|
|X=. .o.=S .|
|Bo .. +o. .|
|.oo .oo o|
|.o .ooo|
|.o .o|
+----[SHA256]-----+
laibaamjad065@ubuntu:~/Assignment01$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAILiwlCjxvMYv1NtoKjFDbvEYHoWm3AcHma87V1SUEH fa24b1-se-021@student.domain
laibaamjad065@ubuntu:~/Assignment01$ git remote set-url github git@github.com:laibaamjad065/Assignment01.git
laibaamjad065@ubuntu:~/Assignment01$ git push -u github main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 252 bytes | 252.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:laibaamjad065/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'github/main'.
laibaamjad065@ubuntu:~/Assignment01$ _

```

2.3 Verify GitHub Repository

Navigated to

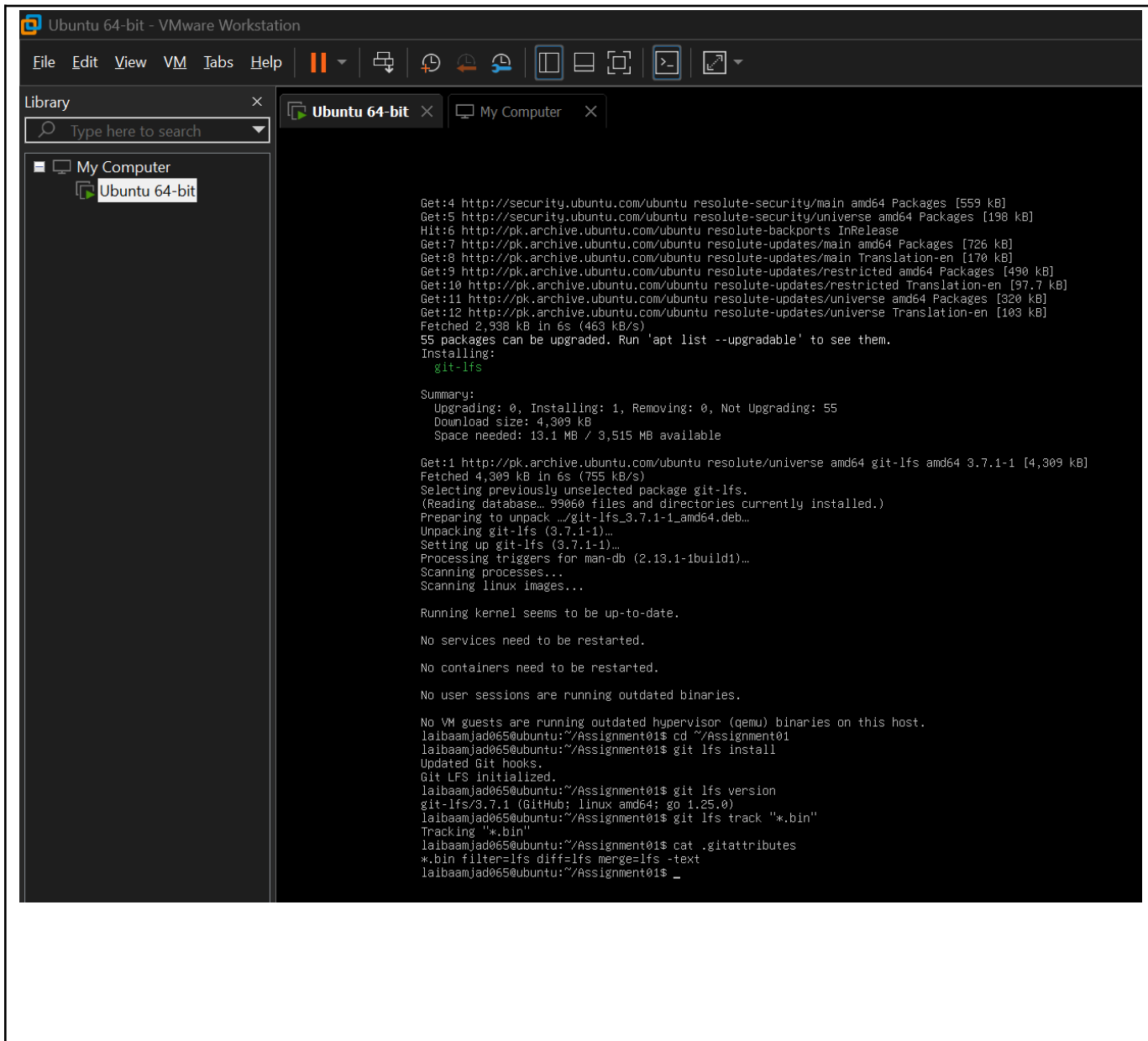
<https://github.com/laibaamjad065/Assignment01> in the web browser to verify that the public repository, owner username, repository name, and rendered README.md file are correctly reflected on GitHub.



TASK 3 — Track Three Large Files with Git LFS

3.1 Configure Git LFS

Initialized Git Large File Storage (Git LFS) inside the **Assignment01** repository and configured tracking for binary (*.bin) files, verifying the generated **.gitattributes** configuration file.



3.2 Create and Verify Three LFS Files

Generated three dummy binary files exceeding 100 MiB each (**large-file-1.bin**, **large-file-2.bin**, **large-file-3.bin**) using the **truncate** utility. Staged the files and verified that Git LFS accurately tracks them along with their designated file sizes.

```
laibaamjad065@ubuntu:~/Assignment01$ cd ~/Assignment01
laibaamjad065@ubuntu:~/Assignment01$ truncate -s 101M large-file-1.bin
laibaamjad065@ubuntu:~/Assignment01$ truncate -s 102M large-file-2.bin
laibaamjad065@ubuntu:~/Assignment01$ truncate -s 103M large-file-3.bin
laibaamjad065@ubuntu:~/Assignment01$ git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin
laibaamjad065@ubuntu:~/Assignment01$ lfs ls-files --size
Command 'lfs' not found, but there are 17 similar ones.
laibaamjad065@ubuntu:~/Assignment01$ git lfs ls-files --size
f1753a034b * large-file-1.bin (106 MB)
2338a970c0 * large-file-2.bin (107 MB)
cb91d5908e * large-file-3.bin (108 MB)
laibaamjad065@ubuntu:~/Assignment01$ _
```

3.3 Commit and Push

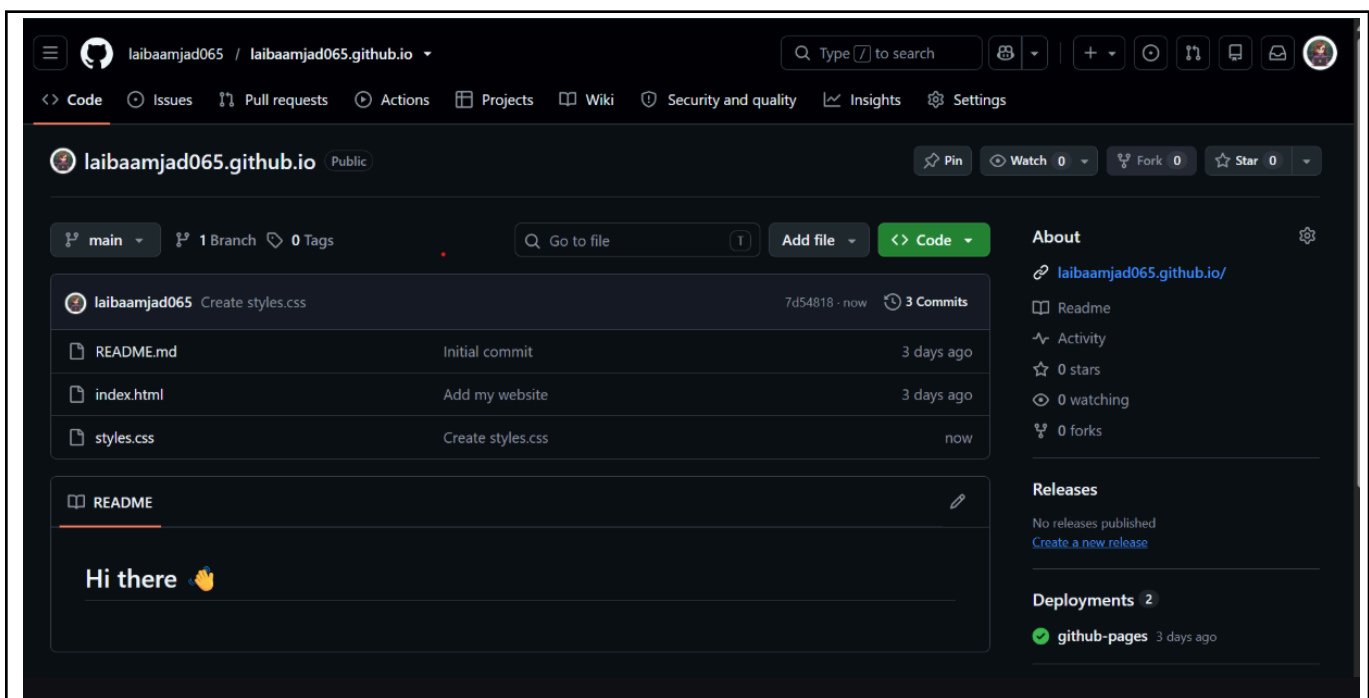
Committed the Git LFS tracking rules and large binary files to the repository, followed by pushing the entire payload to GitHub. Verified that all three LFS objects (total size ~507 MB) uploaded successfully at 100% completion.

```
laibaamjad065@ubuntu:~/Assignment01$ git push https://github.com/laibaamjad065/Assignment01.git main
Username for 'https://github.com': laibaamjad065
Password for 'https://github.com': laibaamjad065
Username for 'https://github.com': laibaamjad065
Password for 'https://github.com': laibaamjad065
Enumerating objects: 7, done.3/3), 507 MB | 430 KB/s
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 802 bytes | 802.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/laibaamjad065/Assignment01.git
 deb6191..5225472 main -> main
laibaamjad065@ubuntu:~/Assignment01$ _
```

TASK 4—Create a Portfolio/CV with GitHub Pages

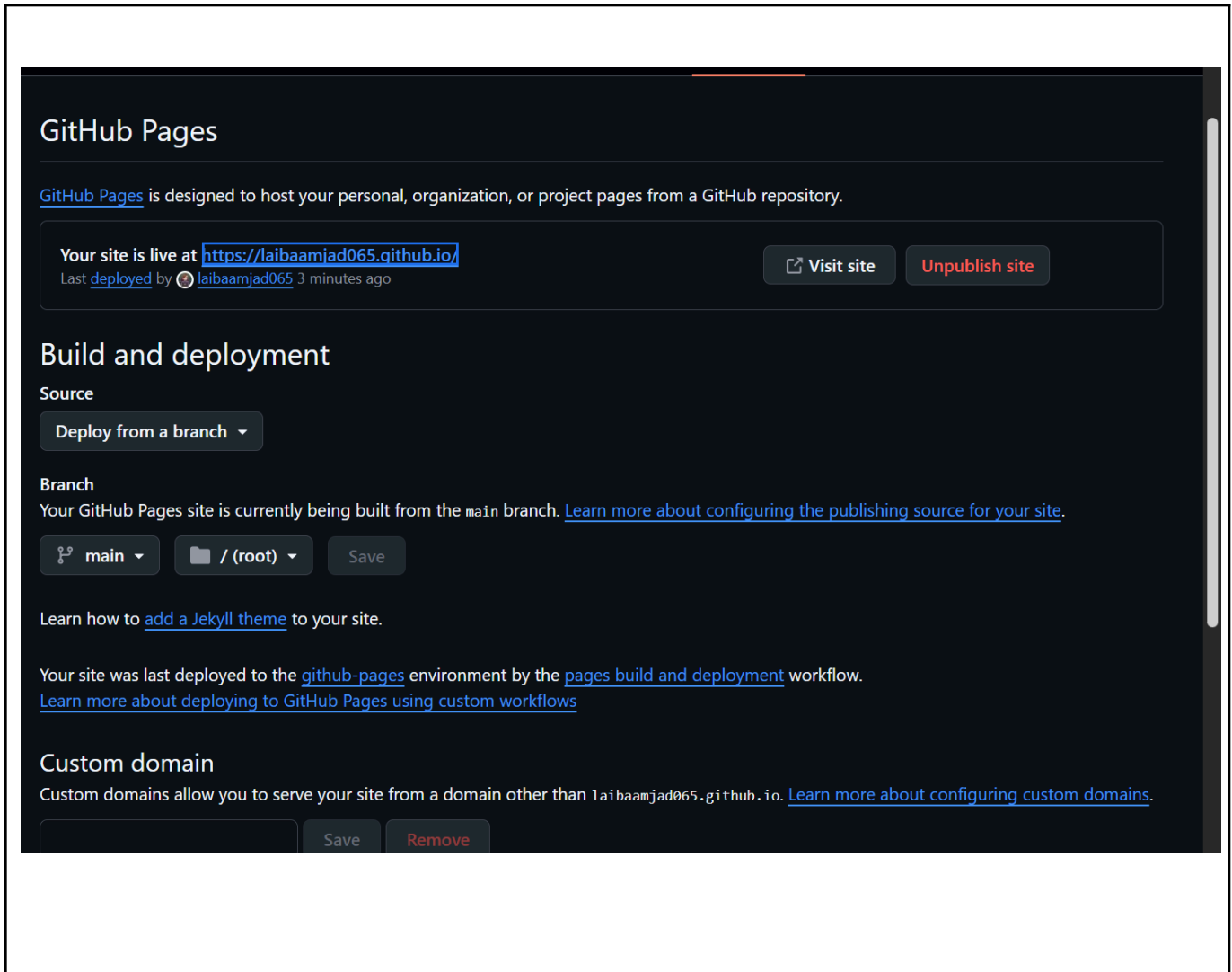
4.1 Portfolio Repository

Configured a dedicated public repository named [laibaamjad065.github.io](https://github.com/laibaamjad065.github.io) containing the necessary web assets ([index.html](#) and [styles.css](#)). Ensured the HTML file incorporates all required sections: **About Me**, **Education**, **Skills**, and **Projects**. Pushed the source code to the [main](#) branch.



4.2 Publish with GitHub Pages

Configured GitHub Pages deployment under **Settings** → **Pages** by assigning the deployment source to the **main** branch and **/ (root)** folder. Confirmed successful build execution indicated by the active status banner: "Your site is live at <https://laibaamjad065.github.io/>".



4.3 Verify Live Portfolio

Accessed <https://laibaamjad065.github.io/> via a web browser to confirm live deployment accessibility. Verified that the website renders properly, displaying the complete URL along with the mandatory sections (About Me, Education, Skills, and Projects).

Laiba Amjad

BS Software Engineering student | Cloud Computing learner

My Skills

- Linux and Git
- Python
- Cloud Computing basics

Email: fa24b1-se-021@student.domain